

10.LAPROSCOPIC TOWER WITH INSTRUMENTS

TECHNICAL SPECIFICATION

4K UHD LAPAROSCOPIC IMAGING TOWER WITH ICG FLUORESCENCE

1. SYSTEM CONFIGURATION

The complete system shall be a high-end integrated **4K UHD surgical imaging platform** suitable for:

- Laparoscopic surgery
- General surgery
- Gastrointestinal surgery
- Colorectal surgery
- Urology
- Gynaecology
- Hepatobiliary surgery
- Bariatric surgery
- Thoracoscopic procedures, where applicable
- ENT endoscopic procedures
- NIR/ICG fluorescence-guided surgery

The system shall be supplied as a complete turnkey solution comprising:

1. 4K UHD camera system
2. 4K camera head
3. 4K surgical monitor
4. LED/advanced surgical light source
5. NIR/ICG fluorescence imaging capability
6. Camera control unit
7. Image management/recording system
8. Medical-grade trolley
9. Insufflator
10. Smoke evacuation system
11. Electrosurgical unit, if included in tender
12. Appropriate laparoscopic/endoscopic instruments
13. ENT endoscopic instruments as listed in the BOQ
14. All required cables, adapters and accessories

2. 4K UHD CAMERA SYSTEM

The camera system shall have:

- Native 4K UHD imaging.
- Minimum resolution of **3840 × 2160 pixels** or better.
- 3-chip CMOS or equivalent high-performance sensor technology.
- Progressive scan imaging.
- High sensitivity and low-light performance.
- Excellent colour reproduction.
- High dynamic range.
- High contrast.
- Low image noise.
- Automatic white balance.
- Manual white balance.
- Automatic brightness control.
- Digital zoom.
- Image enhancement.
- Freeze function.
- Patient data overlay, where applicable.
- Digital image capture.
- Video recording capability.

The camera system shall provide clear visualization of fine anatomical structures and shall be suitable for minimally invasive surgery.

3. 4K CAMERA HEAD

The system shall include:

- 4K UHD camera head.
- Lightweight ergonomic design.
- Sterilizable camera head or sterile drape compatibility.
- Remote camera controls.
- Zoom control.
- Focus control, where applicable.
- White balance control.
- Image capture control.
- Programmable buttons.
- Compatible with all supplied laparoscopic and endoscopic scopes.

The camera head shall provide high-quality 4K images without significant image degradation.

4. 4K SURGICAL MONITOR

The system shall include:

- Medical-grade 4K UHD monitor.
- Minimum **31-inch display preferred**.
- Resolution: Minimum 3840×2160 pixels.
- 16:9 aspect ratio.
- High brightness.
- High contrast.
- Wide viewing angle.
- Anti-glare surface.
- Multiple digital inputs/outputs.
- HDMI/SDI or equivalent professional digital interfaces.
- Picture-in-picture capability preferred.
- Picture-by-picture capability preferred.
- Suitable for operating-room use.

The monitor shall be mounted on the integrated medical-grade equipment trolley or supplied with an appropriate medical-grade monitor stand.

5. NIR / ICG FLUORESCENCE IMAGING

The system shall provide integrated or upgradeable:

Near Infrared Fluorescence Imaging

- NIR fluorescence capability.
- ICG fluorescence imaging.
- Real-time switching between white light and fluorescence mode.
- Visualization of fluorescent anatomical structures.
- Enhanced visualization of blood vessels.
- Biliary anatomy visualization.
- Tissue perfusion assessment.
- Lymphatic visualization, where applicable.
- Suitable for fluorescence-guided minimally invasive surgery.

The system shall allow:

- White light mode.
- NIR/ICG mode.
- Simultaneous or sequential visualization.
- Fluorescence intensity adjustment.
- Fluorescence image enhancement.
- Digital recording of fluorescence images and videos.

6. IMAGE MANAGEMENT SYSTEM

The system shall provide:

- Integrated digital image capture.
- Still image recording.
- Full HD/4K video recording.
- Patient data entry.
- Case data management.
- USB export.
- Network connectivity.
- Digital storage.
- PACS/DICOM compatibility preferred.
- Hospital HIS/EMR integration capability preferred.
- Date/time stamp.
- Patient identification.
- Procedure identification.
- Image annotation.
- Video playback.
- Data export in standard formats.

The system shall have a minimum **1 TB internal storage or better**, or equivalent network storage capability.

7. LIGHT SOURCE

The system shall include:

- High-intensity LED light source.
- Long-life LED illumination.
- Minimum **20,000 hours expected LED life preferred**.
- Adjustable light intensity.
- Automatic light control.
- White light illumination.
- NIR fluorescence excitation capability.
- Low heat generation.
- Automatic standby function.
- Over-temperature protection.

The light source shall be suitable for continuous operation during long surgical procedures.

8. ENDOSCOPIC / LAPAROSCOPIC TELESCOPES

The system shall be compatible with:

- 0° laparoscope.
- 30° laparoscope.
- 45° laparoscope, if required.
- 5 mm laparoscope.
- 10 mm laparoscope.
- 4K-compatible rigid endoscopes.
- NIR/ICG-compatible endoscopes.

The telescope shall have:

- High-resolution optics.
- Wide field of view.
- Excellent light transmission.
- Anti-reflection coating.
- Autoclavable construction.
- Compatible with the proposed 4K camera system.

9. LAPAROSCOPIC INSUFFLATOR

The complete laparoscopic system shall include a high-performance CO₂ insufflator with:

- Electronic CO₂ flow control.
- Automatic pressure regulation.
- Adjustable intra-abdominal pressure.
- High flow capability.
- Minimum flow: **40 L/min or better preferred.**
- Automatic gas leakage compensation.
- Smoke evacuation compatibility.
- Automatic gas heating or heated insufflation preferred.
- Pressure monitoring.
- Audible and visual alarms.
- Over-pressure protection.
- Gas supply monitoring.
- Empty cylinder alarm.

For advanced laparoscopic surgery, a **high-flow insufflator with integrated smoke evacuation** is preferred.

10. SMOKE EVACUATION

The system shall include or be compatible with:

- Integrated surgical smoke evacuation.
- Automatic activation with electrosurgical devices.
- High-efficiency filtration.
- ULPA/HEPA-grade filtration or equivalent.
- Low-noise operation.
- Replaceable filter.
- Visual filter-life indicator.
- Compatible laparoscopic smoke evacuation tubing.

11. ELECTROSURGICAL COMPATIBILITY

The system shall be compatible with:

- Monopolar electrosurgery.
- Bipolar electrosurgery.
- Advanced bipolar devices.
- Vessel sealing devices.
- Ultrasonic energy devices.
- Laparoscopic energy instruments.

Where an electrosurgical unit is included, it should have:

- Monopolar cutting.
- Monopolar coagulation.
- Bipolar coagulation.
- Automatic tissue response.
- Vessel sealing capability.
- Argon plasma coagulation, if required.
- Patient return electrode monitoring.
- Overload protection.
- Audible alarms.

12. MEDICAL-GRADE TROLLEY

The system shall include an integrated mobile trolley with:

- Medical-grade construction.

- Stainless steel or high-grade powder-coated body.
- Minimum 4 castor wheels.
- Lockable wheels.
- Cable management.
- Equipment shelves.
- Lockable storage drawer.
- Monitor mounting system.
- Camera control unit mounting.
- Light source mounting.
- Insufflator mounting.
- Recording system mounting.
- Electrical isolation/medical-grade power distribution.
- Equipotential grounding facility.
- Surge protection.
- UPS support preferred.

13. 4K / NIR-ICG SYSTEM – REQUIRED ACCESSORIES

The complete system shall include:

Sl. No.	Equipment	Qty.
1	4K Camera Control Unit	1
2	4K Camera Head	1
3	4K UHD Medical Monitor	1
4	4K LED Light Source	1
5	NIR/ICG Fluorescence Module	1
6	Image Management/Recording System	1
7	Medical Equipment Trolley	1
8	CO ₂ Insufflator	1
9	Smoke Evacuation System	1
10	4K Compatible 0° Telescope	1
11	4K Compatible 30° Telescope	1
12	NIR/ICG Compatible Telescope	1
13	Camera Coupler	1
14	Light Cable	2
15	CO ₂ Hose	2
16	Sterile Camera Drapes	Initial supply
17	Sterile Light Cable Covers	Initial supply
18	Recording Media/Storage	As required

Sl. No.	Equipment	Qty.
19	Footswitch	1
20	All necessary cables/adapters	1 set

14. ENT ENDOSCOPIC INSTRUMENT SET

The following instruments shall be supplied as part of the **ENT Endoscopic/Laryngeal Instrument Set**.

A. Suction & Elevation Instruments

1. Freer Suction Elevator with stylet, length 19 cm – **1 No.**
2. Suction Elevator, angular, with grip plate and cut-off hole, LUER, distal elevator blade curved to left, delicate tapered sheath, working length 21 cm – **1 No.**
3. Freer Elevator, double-ended, semisharp and blunt, length 20 cm – **1 No.**
4. Cottle Elevator, double-ended, semisharp and blunt, graduated, length 20 cm – **1 No.**
5. Elevator, double-ended, semisharp and blunt, graduated, length 20 cm – **1 No.**
6. Suction Raspatory, working length 23 cm – **1 No.**
7. Suction Tube, OD 2.5 mm, working length 23 cm – **1 No.**

B. Kerrison Bone Punches

8. Kerrison Bone Punch, detachable, rigid, 90° upbiting, non-through-cutting, 1 mm, WL 17 cm – **1 No.**
9. Kerrison Bone Punch, detachable, rigid, 90° upbiting, non-through-cutting, 2 mm, WL 17 cm – **1 No.**
10. Kerrison Bone Punch, detachable, rigid, 90° upbiting, non-through-cutting, 3 mm, WL 17 cm – **1 No.**
11. Kerrison Bone Punch, detachable, rigid, 90° upbiting, non-through-cutting, 4 mm, WL 17 cm – **1 No.**

C. Kleinsasser Micro-Laryngeal Instruments

12. Kleinsasser Needle Holder, delicate, straight, serrated jaws, 1.8 × 3.5 mm, conically reinforced sheath, ratchet, cleaning connector, WL 23 cm – **1 No.**
13. Kleinsasser Laryngeal Biopsy Forceps, heavy model, oval jaws, 3 × 4 mm, single-action jaws, cleaning connector, WL 23 cm – **1 No.**
14. Kleinsasser Scissors, straight, cleaning connector, WL 23 cm – **1 No.**
15. Kleinsasser Scissors, angled 45°, cleaning connector, WL 23 cm – **1 No.**
16. Kleinsasser Scissors, curved right, cleaning connector, WL 23 cm – **1 No.**
17. Kleinsasser Scissors, curved left, cleaning connector, WL 23 cm – **1 No.**

18. Kleinsasser Forceps, straight, 2 mm cupped jaws, cleaning connector, WL 23 cm – **1 No.**
19. Kleinsasser Forceps, curved upwards, 2 mm cupped jaws, cleaning connector, WL 23 cm – **1 No.**
20. Kleinsasser Forceps, curved right, 2 mm cupped jaws, cleaning connector, WL 23 cm – **1 No.**
21. Kleinsasser Forceps, curved left, 2 mm cupped jaws, cleaning connector, WL 23 cm – **1 No.**
22. Kleinsasser Hook, blunt, probe end, WL 23 cm – **1 No.**
23. Kleinsasser Hook, sharp, probe end, WL 23 cm – **1 No.**
24. Kleinsasser Knot Tier, WL 23 cm – **1 No.**
25. Kleinsasser Ligature Needle, curved left, WL 23 cm – **1 No.**
26. Kleinsasser Ligature Needle, curved right, WL 23 cm – **1 No.**
27. Kleinsasser Knife, sickle-shaped, curved, pointed, WL 23 cm – **1 No.**
28. Kleinsasser Knife, golf club-shaped, curved, round, WL 23 cm – **1 No.**
29. Kleinsasser Knife, oval, straight, WL 23 cm – **1 No.**
30. Kleinsasser Suction Raspatory, WL 23 cm – **1 No.**
31. Kleinsasser Suction Tube, OD 2.5 mm, WL 23 cm – **1 No.**
32. Kleinsasser Injection Needle, LUER-Lock, straight, WL 23 cm – **1 No.**
33. Kleinsasser Suction & Coagulation Cannula, OD 3 mm, WL 26 cm – **1 No.**
34. Unipolar High Frequency Cord, 5 mm plug, length 300 cm – **1 No.**

D. Ferguson Suction Tubes

35. Ferguson Suction Tube with cut-off hole and stylet, LUER, 6 Fr., WL 11 cm – **1 No.**
36. Ferguson Suction Tube, 8 Fr., WL 11 cm – **1 No.**
37. Ferguson Suction Tube, 9 Fr., WL 11 cm – **1 No.**
38. Ferguson Suction Tube, 10 Fr., WL 11 cm – **1 No.**
39. Ferguson Suction Tube, 12 Fr., WL 11 cm – **1 No.**
40. Ferguson Suction Tube, 15 Fr., WL 11 cm – **1 No.**

E. Frazier Suction Tubes

41. Frazier Suction Tube with cut-off hole and stylet, angled, 5 Fr./1.5 mm, WL 10 cm – **1 No.**
42. Frazier Suction Tube, 7 Fr./2 mm, WL 10 cm – **1 No.**
43. Frazier Suction Tube, 9 Fr./3 mm, WL 10 cm – **1 No.**

F. Additional ENT / Sinus Instruments

44. Double-ended Curette, oval, one side short curved 45°, semisharp, diameter 2.2 mm; other side long curved 90°, sharp, diameter 2.7 mm, length 19 cm – **1 No.**
45. Maxillary Sinus Ostium Seeker, double-ended, ball ends 1.2 mm and 2 mm, length 19 cm – **1 No.**

15. INSTRUMENT QUALITY REQUIREMENTS

All supplied ENT instruments shall:

- Be manufactured from high-quality surgical-grade stainless steel.
- Be corrosion-resistant.
- Be autoclavable.
- Be reusable.
- Be suitable for hospital CSSD sterilization.
- Have smooth atraumatic working surfaces.
- Have precise jaw alignment.
- Have ergonomic handles.
- Have cleaning connectors wherever specified.
- Be compatible with ultrasonic cleaning where applicable.
- Have laser-etched or permanent identification where required.
- Be supplied with appropriate sterilization trays.

The instruments shall be supplied in suitable **perforated sterilization containers/trays** with instrument identification.

16. INSTRUMENT TRAYS

The bidder shall supply:

- Minimum 2 sterilization trays for ENT instruments.
- Appropriate silicone instrument holders.
- Instrument identification labels.
- Perforated stainless steel or aluminium sterilization trays.
- Suitable sterilization container system.
- Container filters.
- Lid locking mechanism.
- Internal instrument mat.

The quantity of trays shall be sufficient to permit **one tray in use, one tray in sterilization/CSSD cycle**, wherever clinically required.

17. WARRANTY

The complete system shall be supplied with:

- Minimum **3 years comprehensive warranty**.
- Camera system under warranty.
- Monitor under warranty.
- Light source under warranty.
- NIR/ICG module under warranty.
- Insufflator under warranty.
- Recording system warranty.
- Trolley warranty.
- Endoscope warranty.
- Instrument warranty against manufacturing defects.
- Annual calibration, where applicable.
- Minimum 4 preventive maintenance visits per year.
- Breakdown response within 24–48 hours.
- Availability of genuine spare parts for minimum 7–10 years.
- Authorized service support in India.